

E Proofs for Hardness Results

We first review the definition of a group action.

Definition E.1 (group action). Let $(G, \circ)$ be a group with identity e_G and $\mathcal{R}$ be any set. G is said to *act* on $\mathcal{R}$ (from the left) if there exists a map $\cdot : G \times \mathcal{R} \rightarrow \mathcal{R}$ (the *group action*) satisfying the following two axioms:

- (1) (identity) $e_G \cdot r = r$ for all $r \in \mathcal{R}$,
- (2) (composition) $g_1 \cdot (g_2 \cdot r) = (g_1 \circ g_2) \cdot r$ for all $g_1, g_2 \in G$ and $r \in \mathcal{R}$.

It follows that the map $r \mapsto g \cdot r$ is a bijection for all $g \in G$, with inverse $r \mapsto g^{-1} \cdot r$.

E.1 Proof of Theorem 5.2

Let $\mathcal{P}' \subseteq \mathcal{P}$ be such that $|\mathcal{P}'| = \text{SDA}(\mathcal{P}, \mathcal{I})$, h_p are pairwise orthogonal and $\mathcal{I}_p$ are equal for all $p \in \mathcal{P}'$. This ensures that $f_\theta(x, \mathcal{I}_p(x))$ is independent of p . Choosing p uniformly randomly from $\mathcal{P}'$, the variance of the gradient of L with respect to p is computed as

$$\begin{aligned} \text{Var}_p \nabla L(\theta; p) &= \min_{u \in \mathbb{R}} \mathbb{E}_p [(-2\langle \nabla f_\theta, h_p - f_\theta \rangle_{\mathcal{H}} - u)^2] \\ &\leq \mathbb{E}_p [4\langle \nabla f_\theta, h_p \rangle_{\mathcal{H}}^2] \\ &= \frac{4}{|\mathcal{P}'|} \sum_{p \in \mathcal{P}'} \langle \nabla f_\theta, h_p \rangle_{\mathcal{H}}^2 \\ &\leq \frac{4\|\nabla f_\theta\|_{\mathcal{H}}^2}{|\mathcal{P}'|} \leq \delta^3. \end{aligned}$$

It follows from Chebyshev's inequality that for all θ ,

$$\mathbb{P}(\|\nabla L(\theta; p) - \mathbb{E}_p[\nabla L(\theta; p)]\| \geq \delta) \leq \delta.$$

Thus if all queries to ∇L are adversarially corrupted with δ to return $\mathbb{E}_p[\nabla L(\theta; p)]$ when $\|\nabla L(\theta; p) - \mathbb{E}_p[\nabla L(\theta; p)]\| < \delta$, the n successive queries will not reveal any information on the ground truth p with probability $1 - n\delta$ by a union bound. Hence after running the algorithm, the loss is bounded below by the random guessing error with any fixed θ . Noting that replacing f_θ with $\bar{f}_\theta(x) := \max\{\min\{f_\theta(x), 1\}, -1\}$ does not increase the loss, it follows that

$$\begin{aligned} \mathbb{E}_p [L(\theta; p)] &\geq \mathbb{E}_p [\|h_p - \bar{f}_\theta\|_{\mathcal{H}}^2] \\ &= 1 + \|\bar{f}_\theta\|_{\mathcal{H}}^2 - 2\mathbb{E}_p [\langle h_p, \bar{f}_\theta \rangle_{\mathcal{H}}] \\ &= 1 + \|\bar{f}_\theta\|_{\mathcal{H}}^2 - \frac{2}{|\mathcal{P}'|} \left\langle \sum_{p \in \mathcal{P}'} h_p, \bar{f}_\theta \right\rangle_{\mathcal{H}} \end{aligned}$$

$$\begin{aligned}
&= 1 + \left\| \bar{f}_\theta - \frac{1}{|\mathcal{P}'|} \sum_{p \in \mathcal{P}'} h_p \right\|_{\mathcal{H}}^2 - \frac{1}{|\mathcal{P}'|^2} \sum_{p, p' \in \mathcal{P}'} \langle h_p, h_{p'} \rangle_{\mathcal{H}} \\
&\geq 1 - \frac{1}{|\mathcal{P}'|},
\end{aligned}$$

uniformly for all θ . □

E.2 Proof of Theorem 5.3

Note that $\text{SDA}(\mathcal{P}; \mathcal{P}) = 1$ is trivial. We give three constructions realizing the lower bounds in order of strictness of the additional information constraint.

Lower bounding $\text{SDA}(\mathcal{P}; \emptyset)$. Suppose $p = p^\varepsilon$ is any kernel satisfying Assumptions 1, 2 and $\mathcal{D}$ is any distribution satisfying Assumption 3'. We first analyze the case where the lower bound of Assumption 3' is $\Omega(\log K)$ and $M \geq \Omega(K)$, since the proof is slightly more involved. Let $s_{k,1}, \dots, s_{k,|C_k|}$ be a labeling of all states in C_k such that all nodes with outbound sparse edges are contained in the first n_{out} nodes. Define the quantity

$$q := \left\lfloor \frac{\min_{k \in K} |C_k|}{n_{\text{out}}} \right\rfloor = \Theta(M).$$

For a vector $v \in \mathbb{Z}_q^K$, define the corresponding permutation of S (also denoted by v) as

$$v(s_{k,i}) = \begin{cases} s_{k, (i + v_k n_{\text{out}} - 1 \bmod q n_{\text{out}}) + 1} & 1 \leq i \leq q n_{\text{out}} \\ s_{k,i} & q n_{\text{out}} < i \leq |C_k|. \end{cases}$$

That is, the first $q n_{\text{out}}$ states of C_k are cyclically shifted by v_k -multiples of n_{out} and the remaining states are left fixed. Denote the pushforward kernel of p^ε induced by v as

$$v_\# p^\varepsilon(y|x) = p^\varepsilon(v^{-1}(y)|v^{-1}(x)).$$

It is clear that $v_\# p^\varepsilon \in \mathcal{P}$ for all v , moreover, Assumptions 1, 2 hold when replacing E_s by $E_s(v_\# p^\varepsilon)$ since v only permutes states within clusters and does not affect the sparse structure.

Lemma E.2. *Let d_H denote the Hamming distance on $\mathbb{Z}_q^K$. For any two $v, v' \in \mathbb{Z}_q^K$ it holds that*

$$|E_s(v_\# p^\varepsilon)| \leq n_{\text{out}} d_{\text{out}} K$$

and

$$|E_s(v_\# p^\varepsilon) \cap E_s(v'_\# p^\varepsilon)| \leq n_{\text{out}} d_{\text{out}} (K - d_H(v, v')).$$

Proof. Suppose for some $x \in C_k, y \in C_\ell$ with $k \neq \ell$ we have $(x, y) \in E_s(v_\# p^\varepsilon)$. Then $(v^{-1}(x), v^{-1}(y)) \in E_s(p^\varepsilon)$ so that $v^{-1}(x) = s_{k,i}$ for some $1 \leq i \leq n_{\text{out}}$, hence $x = v(s_{k,i}) = s_{k,i+v_k n_{\text{out}}}$. If at the same time $(x, y) \in E_s(v'_\# p^\varepsilon)$, it must hold that $s_{k,i+v_k n_{\text{out}}} = s_{k,i+v'_k n_{\text{out}}}$ and so k must satisfy $v_k = v'_k$. There are exactly $K - d_H(v, v')$ such clusters and at most $n_{\text{out}} d_{\text{out}}$ sparse edges leading out of each cluster, concluding the second bound. The first bound follows from the second by setting $v = v'$. $\square$

We construct a well-separated subset of $\mathbb{Z}_q^K$ via the Gilbert-Varshamov bound.

Lemma E.3 (Gilbert-Varshamov bound). *The maximum size $A_q(K, d)$ of a code of length K over an alphabet of size q with minimum Hamming distance d satisfies*

$$A_q(K, d) \geq \frac{q^K}{\text{Vol}_q(K, d-1)}, \quad \text{Vol}_q(K, d) = \sum_{i=0}^d \binom{K}{i} (q-1)^i.$$

Moreover for $\tau \in [0, 1 - 1/q]$ it holds that

$$\text{Vol}_q(K, \tau K) \leq q^{H_q(\tau)K},$$

where H_q is the q -ary entropy function

$$H_q(\tau) = \tau \log_q(q-1) - \tau \log_q \tau - (1-\tau) \log_q(1-\tau).$$

While the classical bound only guarantees the existence of large subsets with linear (at least K/q) overlapping bits, we can obtain a better separation by scaling q (equivalently, M) along with K . In particular, choosing the relative overlap as $O(\log q/q)$ (note the use of the natural logarithm), we obtain:

Lemma E.4. *For $C > 0$, $\tau = 1 - \log q/(Cq)$ and sufficiently large q , it holds that*

$$A_q(K, \tau K) \geq q^{\frac{\log \log q}{2Cq} K}. \quad (21)$$

Proof. Using the inequality $0 \leq x - \log(1+x) \leq x^2$ for $|x| \leq 1/2$, we may bound

$$\begin{aligned} H_q(\tau) &= \left(1 - \frac{\log q}{Cq}\right) \log_q(q-1) - \left(1 - \frac{\log q}{Cq}\right) \log_q \left(1 - \frac{\log q}{Cq}\right) - \frac{\log q}{Cq} \log_q \frac{\log q}{Cq} \\ &= 1 - \frac{\log \log q}{Cq} + \frac{\log C}{Cq} + \left(\frac{1}{\log q} - \frac{1}{Cq}\right) \left(\log \left(1 - \frac{1}{q}\right) - \log \left(1 - \frac{\log q}{Cq}\right)\right) \\ &\leq 1 - \frac{\log \log q}{Cq} + \frac{\log C}{Cq} + \frac{1}{\log q} \left(-\frac{1}{q} + \frac{\log q}{Cq} + \frac{(\log q)^2}{C^2 q^2}\right) \\ &\leq 1 - \frac{\log \log q}{2Cq} \end{aligned}$$

for sufficiently large q . The statement follows from Lemma E.3. $\square$